

TABLE OF CONTENTS
S-210 FIELD AND LABORATORY TEST

	Page No.
1. SCOPE OF WORK	S.210-2
2. CODES AND STANDARDS	S.210-2
3. TESTING AND MONITORING.....	S.210-2
3.1 General.....	S.210-2
3.2 Testing	S.210-2
4. TESTING FREQUENCY.....	S.210-3
4.1 Frequency.....	S.210-3
4.2 Fill Material- Granular.....	S.210-3
4.3 Fill Material- Cohesive	S.210-5
5. TEST RECORDS.....	S.210-6

S.210 FIELD AND LABORATORY TEST

1. SCOPE OF WORK

The Bidder shall furnish the services of a testing laboratory to perform all tests of piles, earth work, structural steel, welding, concrete, and other tests as specified herein. The services of the testing laboratory shall include furnishing all labour, technicians, materials, testing equipment/apparatus as required or as requested by the owner, record keeping and submittal of records of testing to the Owner.

2. CODES AND STANDARDS

The Bidder shall comply with the requirement of Part 4.

The Codes and Standards of the following organizations extracted from the Part 4 are specifically applicable to the design, manufacture and testing of the work included in this Specification.

ASTM	American Society for Testing and Materials
AASHTO	American Association of State Highway and Transportation Officials
COE	U.S. Army Corps of Engineers
JIS	Japanese Industrial Standards

3. TESTING AND MONITORING

3.1 General

The Bidder shall furnish the testing laboratory as required for testing and inspection.

Bidder shall submit with his Bid Proposal, the name and location of the testing laboratory with past experience records of the company and curriculum vitae of the responsible engineers and technicians who will be assigned to this Project.

3.2 Testing

The tests for excavation, fill and soil properties shall be performed in accordance the Technical Specification S-300 Earth with ASTM and/or AASHTO standards or as directed by the Owner. Tests and data recording shall include the following:

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|---|--|
| 1) Classification of soils: | ASTM D2487 and ASTM D2488 |
| 2) Moisture content of soil: | ASTM D2216 |
| 3) Unit weight of soil: | EM-110-2-1906, U.S. Army Corps of Engineers |
| 4) Liquid limit of soils: | ASTM D423 |
| 5) Plastic limit and plasticity index of soil: | ASTM D424 |
| 6) Particle size analysis of soil: | ASTM D422 |
| 7) Amount of material in soil finer than No. 200 sieve: | ASTM D1140 |
| 8) Unconsolidated, undrained compressive strength of cohesive soil in tri-axial compression: | EM-1110-2-1906, U.S. Army Corps of Engineers |
| 9) One dimensional consolidation properties: | ASTM D2435 |
| 10) Permeability test: | EM-110-2-1906, U.S. Army Corps of Engineer |
| 11) Moisture density relation of soil and soil aggregate mixture using 10 lb (4.54 kg) rammer and 12 inch (302mm) drop: | ASTM D698 |
| 12) Moisture density relation of soil and soil aggregate mixture using 10 lb (4.54 kg) rammer and 18 inch (453mm) drop: | ASTM D1557 |
| 13) Compressive strength, unconfined, of cohesive soil: | ASTM D2166 |

4. TESTING FREQUENCY

4.1 Frequency

The testing frequency for soil investigation, piles, concrete, steel structure, welding, etc. shall be as specified in the respective Technical Specification.

The testing frequencies for earthwork given below are approximate only. Depending on the field conditions, the owner may revise, add or delete tests at no additional cost to the Owner.

4.2 Fill Material- Granular

Prior to the start of fill placement, the following tests shall be performed to determine the initial soil properties for the granular fill having a maximum size of 3 inches (75mm). The number of tests given is the minimum required. If the test results are inconsistent, additional tests shall be performed.

<u>Test</u>	<u>ASTM or AASHTO</u>	<u>Number</u>
1. Classification of soils	D2487 or D2488	5
2. Moisture content	D2216	5
3. Particle size	D422	5
4. Specific gravity of soil	D854	5
5. UU Triaxial test	D2850	5
6. Direct shear test	D3080	5
7. Moisture density relation	D698/ D1557	5

If, during course of work, the material used as granular fill changes, additional tests shall be performed to reclassify the material.

During fill placement, the following tests shall be performed to verify that the material is the same as initially defined and to verify compliance with the in-place density requirements as specified in the Technical Specification- S300 Earth Work.

<u>Laboratory tests</u>	<u>ASTM or AASHTO</u>	<u>Frequency</u>
1. Classification of soil	D2487 or D2488	one/5000m ³
2. Moisture content	D2216	one/5000m ³
3. Particle size sieve	D422	one/5000m ³
4. Relative density	D2049	one/5000m ³
5. Moisture density relation	D698/D1557	one/5000m ³

<u>Field tests</u>	<u>ASTM or AASHTO</u>	<u>Frequency</u>
1. Moisture content	D2216	one/each in-place density test
2. In-place density test- general	D1556	one/1000m ³
3. In-place density test- structural	D1556	one/200m ³ but not less than one per lift
4. In-place density-trenches	D1556	one/75 lineal meters of trench but not less than one per lift
5. Lift thickness		one/lift

4.3 Fill Material- Cohesive

If cohesive material is used as fill material, the following tests shall be performed to determine the initial soil properties for the cohesive fill prior to the start of fill placement. The number of tests given is the minimum required. If the test results are inconsistent, additional tests shall be performed, without additional cost.

<u>Tests</u>	<u>ASTM or AASHTO</u>	<u>Number</u>
1. Classification of soil	D2487 or D2488	5
2. Moisture content	D2216	5
3. Particle size	D422	5
4. Liquid limit	D423	5
5. Plastic limit	D424	5
6. Specific gravity of soil	D854	5
7. UU triaxial test	D2850	5
8. CU triaxial test	“Lambe”	5
9. Consolidation tests	D2435	5
10. Unconfined compressive strength	D2166	5
11. Direct shear test	D3080	5
12. Moisture density relation	D698/D1577	5

If, during the course of work, the material used as cohesive fill changes, additional tests shall be performed to reclassify the material without additional cost to the Owner.

During fill placement, the following tests shall be performed to verify that the material is the same as initially defined and to verify compliance with the in-place density requirements as specified in the Technical Specification- S300 Earth Work.

<u>Laboratory Test</u>	<u>ASTM or AASHTO</u>	<u>Frequency</u>
1. Classification of soils	D2487 or D2488	one/5000m ³
2. Moisture content	D2216	one/5000m ³
3. Particle size-sieve analysis	D422	one/5000m ³
4. Liquid limit	D423	one/5000m ³
5. Plastic limit	D242	one/5000m ³
6. Moisture density relation	D698/ D1557	one/5000m ³

<u>Field tests</u>	<u>ASTM or AASHTO</u>	<u>Frequency</u>
1. Moisture content	D2216	one/each in-place density test
2. In-place density test- general and random	D2167	one/1000m ³
3. In-place density – structural fill	D2167	one/200m ³ but not less than one per lift
4. In-place density-trenches	D2167	one/75 lineal meters of trench but not less than one per lift
5. Lift thickness		one/lift

5. TEST RECORDS

The Bidder shall furnish copies of all test results in accordance with the Technical Specification and in accordance with the agreed upon schedule for the Owner's review and records. Prior to the use of materials, the Bidder shall submit copies of the test records for approval by the owner at least two weeks before the materials is to be used for construction. If in the opinion of the Owner, any test result is of questionable quality, the Bidder shall perform additional tests as directed by the Owner at no cost to the Owner.

(Last page of S-210)